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Research on influence of steam extraction parameters and operation load on operational flexibility of coal-fired power plant

Haijiao Wei [a], Yuanwei Lu [a,*], Yanchun Yang [a], Cancan Zhang [a], Cong He [a], Yuting Wu [a], Weidong Li [b], Dongming Zhao [c]

[a] MOE Key Laboratory of Enhanced Heat Transfer and Energy Conservation, Beijing Key Laboratory of Heat Transfer and Energy Conversion, Faculty of Environment and Life, Beijing University of Technology, Beijing 100124, China
[b] China Huaneng Group Clean Energy Research Institute Co. Ltd, Beijing 102209, China
[c] China Huaneng Group Co. Ltd, Beijing 100031, China

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ABSTRACT

The operational flexibility of coal-fired power plant is very important for the integration of large-scale renewable energy to the grid. In order to increase the operational flexibility of coal-fired power plant, a 600 MW subcritical coal-fired power plant was selected as research example to analyze the influence of steam extraction parameters and operation load by simulation. A novel main steam extraction system was proposed to adjust the power output under the low load period. The sensible and latent heat of the extracted steam was stored in molten salts and phase change materials, respectively, during the load reduction process. While the stored heat was reused again to heat the feed water and condensate water during the load raising process. The 50% rated load (300.03 MW) of the unit was used as the research benchmark to further adjust the load. The results show that the proposed steam extraction system can improve the operational flexibility of coal-fired power plant. The power output can be reduced to 204.51 MW from 300.03 MW when the main steam extraction mass flow rate is 250 t/h during the load reduction process. The sensible heat storage power factor and the operation load of the unit can affect the power output increment of coal-fired power plant during the load raising process. Under the condition of the maximal sensible heat storage power factor is 0.2462, the power output can be increased to 325.47 MW from 300.03 MW when the bypassed feed water mass flow rate of 251 t/h was heated by the stored heat during the load raising process. Using such operation model, the coal-fired power plant can achieve the maximum deep peak shaving time of 8.4 h/d, and its equivalent thermal efficiency is increased by 2.28%. With the same stored thermal energy to heat the bypassed feed water of 308.65 t/h, the power output can be increased to 394.2 MW from 360.07 MW (60% rated load) at the same sensible heat storage power factor of 0.2462 during the load raising process, and the coal-fired power plant can achieve the maximum deep peak shaving time of 9.86 h/d with its equivalent thermal efficiency increase of 2.55%. The results show that the proposed steam extraction method can make the unit operate in the lowest stable load and provide a wider renewable energy access space below the minimum stable load. The research provides a theoretical guidance for the flexibility design of coal-fired power plant, especially suitable for the large-scale renewable energy access to the grid in China.

1. Introduction

The environmental problems caused by carbon emissions can be effectively solved by reducing the consumption of fossil energy [1,2]. The development of renewable energy is an inevitable choice for different countries, such as in China, the 3060 double carbon target has been proposed to formulate the future energy policies. Developing the renewable energy will be the future energy utilization direction in China. However, the large-scale renewable energy (e.g. wind and solar energy) has the characteristics of intermittency and instability [3], which directly affects the safety of the grid operation [4,5]. The operational flexibility of coal-fired power plant (CFPP) [6] should be proposed for the future power grid regulation. The function of the CFPP has to change from the main power supply mode to the peak load regulation mode to make more and more renewable energy power access to the grid. There are two approaches to achieve the operational flexibility of CFPP. One is improving the load response rate [7], the other is adjusting

* Corresponding author.
E-mail address: luyuanwei@bjut.edu.cn (Y. Lu).

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Nomenclature

Abbreviations

THA	turbine heat acceptance
CFPP	coal-fired power plant
TES	thermal energy storage
TER	thermal energy release
HP	high pressure turbine
IP	intermediate pressure turbine
LP	low pressure turbine
RH	regenerative heat exchanger
CP	condensate water pump
FP	feed water pump
TV	throttling valve
SHSHE	sensible heat storage heat exchanger
SHRHE	sensible heat release heat exchanger
PCMHE	phase change material heat exchanger
CT	cold tank
HT	hot tank

Mathematical Symbols

Q	heat power, MW
W	power, MW
G	mass flow rate, t/h
h	specific enthalphy, kJ/kg
q	enthalpy difference of steam, kJ/kg
p	pressure, MPa
T	thermodynamic temperature, K

Greek Symbols

τ	time, h
γ	enthalpy difference of drainage water, kJ/kg
φ	enthalpy difference of water, kJ/kg
η	thermal efficiency, %
ψ	lower heating value of coal, kJ/kg
ε	coefficient of heat preservation, %
θ	sensible heat storage power factor

Subscripts

T	steam turbine
Ts	stage of steam turbine
d	drainage water
w	water
f	feed water
c	condensate water
f,by	bypassed feed water
c,by	bypassed condensate water
sd	drainage water of extracted steam
s	main steam
r	reheat steam
e	exhaust steam of low pressure turbine
st	process of thermal energy storage
re	process of thermal energy release
dt	dividing temperature of extracted steam
b	boiler
pu	pump
0	parameters before off-design condition

the unit load range [8].

Adjusting the load response rate of CFPP was proposed to achieve the operational flexibility of CFPP. Wang et al. [9] realized the operational flexibility of CFPP at the range of 50–100% rated load by changing the water coal ratio. Zhou et al. [10] added a throttle valve in the extraction system of regenerative heater of the CFPP and proposed an improved coordinated control technology to realize the improvement of dynamic response ability and the operational flexibility of variable operation. Wang et al. [11,12] changed successively the control logic of feed water bypass system and the coordinated boiler turbine heating mode. The load response rate and capacity of the unit were increased by 2 times and 2.9% rated power, respectively. The above studies show that the operational flexibility of CFPP can be improved by adjusting the load response. However, adjusting the load response rate of CFPP is only suitable for accepting more unstable renewable energy, and cannot provide more space for renewable energy.

The utilization of thermal energy storage (TES) can realize the highly realizable decouple of the power generation in space and time [13,14], and improve the operational flexibility of CFPP to give more space for renewable energy. Lai et al. [15] added the hot water storage system into the cogeneration unit. The exhaust steam of the intermediate pressure turbine was extracted to heat the stored hot water. When the heat load is 300 MW, the unit load can be continued to decrease by 12%. Richter et al. [16] proposed that added the steam extraction system and the hot water storage devices into thermal power plant to reduce the unit load. The unit load was decreased by 7% in the TES process, and increased by 4.3% in the thermal energy release (TER) process. Trojan et al. [17] stored the pressured water of deaerator in hot water storage tank during the low load period. The unit load was decreased by 21.96 MW, and the lower load of the unit can be reached 67.5%. Li et al. [18] investigated the flexible regulation capacity of 600 MW supercritical CFPP by using TES of phase change materials. The unit load can be decreased by 13.3% and increased by 7.4% under the rated condition, respectively. Cao et al. [19] studied that the excessively electrical power of the unit was used to heat the molten salt under the rated condition. The stored thermal energy of molten salt was used to heat water to generate steam and to drive the auxiliary steam turbine. During the TES and TER process, the unit load was decreased by 16.67% and increased by 6.1% respectively. The previous researches on the operational flexibility of CFPP is only studied under the high load or the rated load. There is no literature on the operational flexibility characteristics of CFPP under low load. If the large-scale renewable energy is access to the grid, the power out of CFPP should be reduced lower, and even lower than the minimum stable load of the units. Thus, the deep peak shaving of CFPP should be urgently proposed to meet the demand of continuous load reduction. In addition, the excessive heat is stored in pressurized water and the generated surplus power is transferred to heat, and the stored heat is reused in the peak load process. Which results in the lower energy efficiency for the low heat density of pressure water, and for the heat from coal transferred to electricity and electricity transferred to heat again. So, the energy conversion efficiency of TES should be improved.

In addition, due to the difference between the coal used in the actual operation and design operation, the minimum operating load of the unit should not be less than 50% turbine heat acceptance condition (THA). The extracted steam with the higher temperature has the great vapor sensible heat and latent heat. If the total heat of the extracted steam is stored by sensible material, which has the disadvantages of pinch point temperature difference [20] and large latent heat of steam condensation [21], so that the temperature of thermal storage material is only about steam saturation temperature. If the total heat of extracted steam is stored by a kind of phase change material, the temperature of phase change material is not higher than the steam saturation temperature, and higher steam temperature is easy to cause the decomposition of phase change material. The higher temperature of the working medium results the higher energy efficiency in CFPP. So, the sensible and latent heat of extracted steam is stored in sensible material and phase change material, respectively, which will improve the energy efficiency of the CFPP.

Therefore, the unit load of 50 %THA is chosen as minimum stable load to reduce the power output through the main steam extraction, and the heat of extracted steam is stored in molten salts and phase change materials, respectively. The stored heat is used to heat the bypassed feed water and condensate water to reduce the steam extraction of regenerate heaters, so that the power output is increased.

The innovation of this research is to use the main steam extraction system to reduce the power output of the CFPP under the minimum stable load, and the heat of the extracted steam is stored and used again, so that its thermal energy efficiency is higher. The advantage of such steam extraction is that on the one hand, the boiler may operate in the lowest stable combustion load, and on the other hand, the power output of CFPP can be lowed below the minimum stable load and provide more space for renewable energy to the grid. The main research of this manuscript is to investigate the feasibility of operational flexibility of CFPP using main steam extraction to reduce the minimum operating load, and analyze the parameters influences of the extracted steam and the operation load on its operation range. The influences of the heat ratio of sensible heat to the total heat of extracted steam and the operation load of unit on the operational flexibility were studied and the energy efficiency was analyzed.

2. Studied system

The 600 MW subcritical CFPP was selected as a research object. The power output of CFPP is 600.18 MW at 100% THA.

2.1. Model of traditional CFPP

Fig. 1 shows the thermal system diagram of the 600 MW subcritical CFPP. The main steam from the boiler enters into high pressure turbine (HP). The exhaust steam of HP enters into the boiler for reheating. The reheated steam enters into intermediate pressure turbine (IP) and low pressure turbine (LP) successively. The exhaust steam of LP condenses in condenser, and the condensate water (The pressured water in 5th-7th regenerative heaters is called condensate water.) pressurized by the condensate pump (CP) enters into 7th regenerative heater (RH7), RH6, RH5 and RH4 successively. The feed water (The pressured water in 1st-3rd regenerative heaters is called feed water.) in RH4 further pressurized by feed water pump (FP) enters into RH3, RH2 and RH1 successively. The feed water is heated in boiler to generate main steam, thus completing a thermodynamic cycle.

The initial parameters of each state points of CFPP under 100% THA are shown in Table 1.

2.2. Operational flexibility model of CFPP

Fig. 2 shows the operational flexibility schematic diagram of CFPP in the load reduction and raising.

The main steam extraction system is integrated with CFPP. The unit load can be reduced by reducing the steam intake of steam turbine during the load reduction process (red line). The sensible heat of the extracted steam is stored in molten salts (e.g. KNO_3-$NaNO_3$-$LiNO_3$-$Ca(NO_3)_2{\cdot}4H_2O$ [22], temperature range 363.15–873.15 K), and the latent heat of the extracted steam is stored in phase change materials (e.g. Xylose-D [23], melting temperature 420.15–424.15 K and latent heat 216–280 kJ/kg). The main steam was extracted from the pipe (point A) and isenthalpic depressurized in throttling valve (TV). The depressurized steam enters the sensible heat storage heat exchanger (SHSHE). At the same time, the low temperature molten salt in the cold tank (CT) is pumped out to the SHSHE, and heated by the extracted steam. The temperature of molten salts rise and then enter into hot tank (HT). After the process of heat exchange between extracted steam and molten salts, the cooled steam enters the phase change material heat exchanger (PCMHE) to continue heating the phase change materials. The extracted steam condenses in PCMHE and flows back to RH4 in the form of cooling

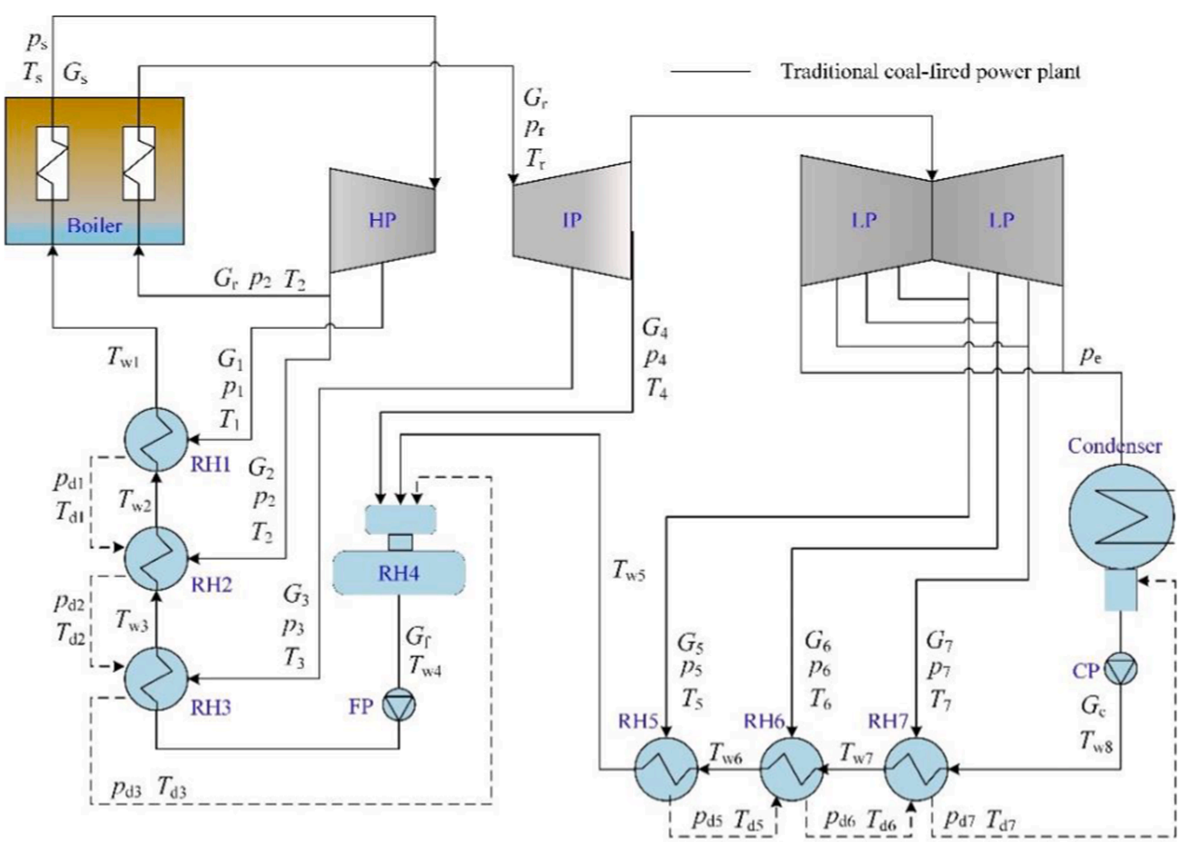

Fig. 1. Thermal system of 600 MW CFPP.

Table 1
Parameters of each state points of the CFPP under 100% THA.

State point	p (MPa)	T (K)	G (t/h)	State point	p (MPa)	T (K)	G (t/h)
s	16.670	811.15	1848.8	w4	1.016	451.55	1848.8
r	3.414	811.15	1576.13	w5	1.724	428.55	1457.65
1	6.081	658.45	131.95	w6	1.724	394.85	1457.65
2	3.794	595.65	127.53	w7	1.724	362.70	1457.65
3	2.046	734.65	69.37	w8	0.015	327.15	1457.65
4	1.016	634.15	54.4	dw1	5.900	524.15	131.95
5	0.615	575.65	83.07	dw2	3.680	489.65	259.48
6	0.240	469.45	75.66	dw3	1.980	460.75	328.85
7	0.081	366.88	79.2	dw5	0.584	400.45	83.07
w1	20.154	549.15	1848.8	dw6	0.228	368.30	158.73
w2	20.154	518.55	1848.8	dw7	0.077	333.55	237.93
w3	20.154	484.05	1848.8	e	0.015	327.15	1218.29

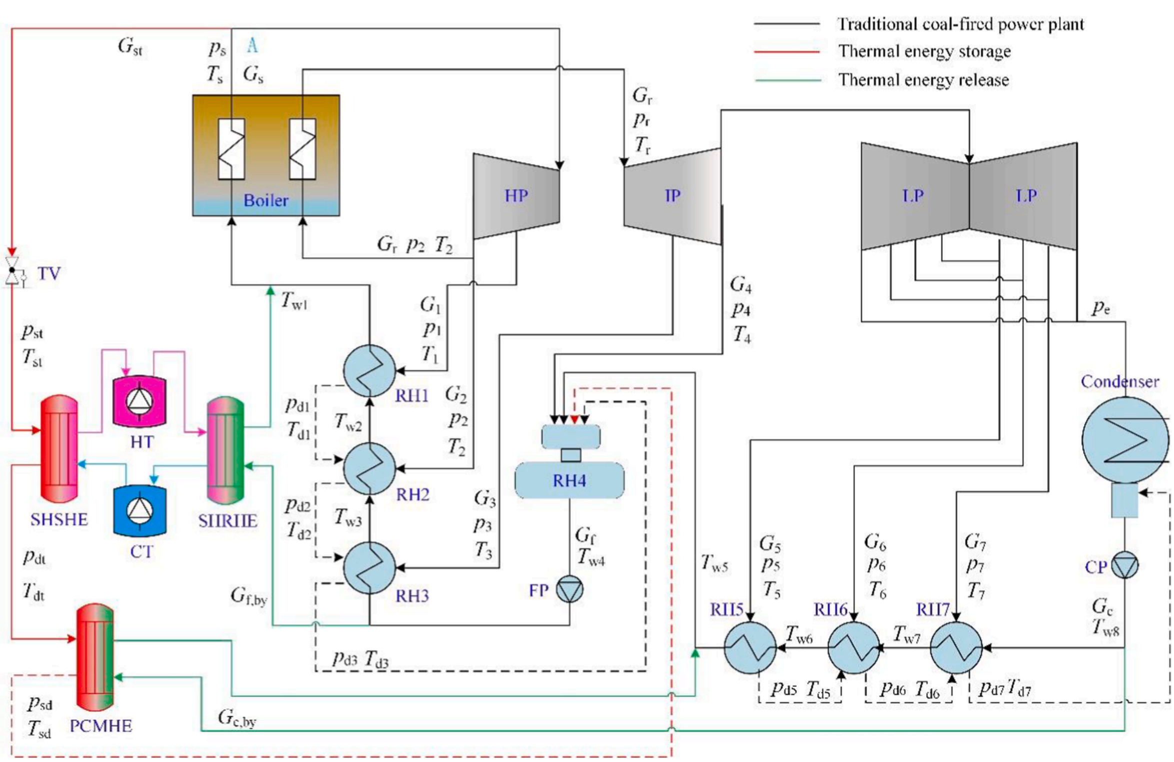

Fig. 2. Operational flexibility schematic diagram of CFPP.

water. The operation mode of CFPP remains unchanged in load reduction process.

The stored heat of the extracted steam is reused to heat the feed water and the condensate water, respectively, during the load raising process (green line). The inter-stage steam extraction of the steam turbine reduces and the power output of the CFPP increases. The high temperature molten salts in the HT is pumped out to the sensible heat release heat exchanger (SHRHE) to heat the feed water. The feed water is heated and enters into the boiler. The cooled molten salts returns to CT. At the same time, the condensate water from condenser enters into the PCMHE and is heated by phase change materials. After the process of heat exchange between condensate water and phase change materials, the high temperature condensate water enters into RH4. The operation mode of CFPP also remains unchanged in load raising process.

3. Methodology for operational flexibility of CFPP

In this paper, the first law of thermodynamics is used to calculate the thermodynamic parameters of operational flexibility system of CFPP.

3.1. Heat balance equation of traditional CFPP

The traditional heat balance calculation of CFPP mainly focuses on the relationship among the enthalpy difference of extracted steam, pressured water and drainage water in RH1-RH7. It can be calculated according to Eq. (1) [24].

$$\begin{bmatrix} q_1 & & & & & & \\ \gamma_2 & q_2 & & & & & \\ \gamma_3 & \gamma_3 & q_3 & & & & \\ \gamma_4 & \gamma_4 & \gamma_4 & q_4 & & & \\ & & & & q_5 & & \\ & & & & \gamma_6 & q_6 & \\ & & & & \gamma_7 & \gamma_7 & q_7 \end{bmatrix} \begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \\ G_6 \\ G_7 \end{bmatrix} = \begin{bmatrix} G_f\varphi_1 \\ G_f\varphi_2 \\ G_f\varphi_3 \\ G_c\varphi_4 \\ G_c\varphi_5 \\ G_c\varphi_6 \\ G_c\varphi_7 \end{bmatrix} \quad (1)$$

The enthalpy difference of extracted steam, pressured water and drainage water is calculated according to Eqs. (2)–(4) [24].

$$q_j = h_j - h_{dj} \quad (2)$$

$$\gamma_j = h_{dj-1} - h_{dj} \quad (3)$$

$$\varphi_j = h_{wj} - h_{wj+1} \tag{4}$$

where, j is the regenerative heater number ($j = 1,2,\cdots,7$).

The power output of steam turbine is obtained according to the relationship between the mass flow rate of main steam and extracted steam. As shown in Eq. (5).

$$W_T = G_s(h_s - h_1) + (G_s - G_1)(h_1 - h_2) + (G_s - G_1 - G_2)(h_r - h_3) + \sum_{m=3}^{6}\left(\left(G_s - \sum_{j=1}^{m} G_j\right)(h_m - h_{m+1})\right) + \left(G_s - \sum_{j=1}^{7} G_j\right)(h_7 - h_e) \tag{5}$$

3.2. Heat balance equation of load reduction and raising process

The sensible heat and latent heat of the extracted steam is stored in molten salts and phase change materials, respectively. The heat storage power of the SHSHE and the PCMHE, and the total heat storage power of extracted steam are calculated according to Eqs. (6)–(8).

$$Q_{st.SHSHE} = G_{st}(h_s - h_{dt}) \tag{6}$$

$$Q_{st.PCMHE} = G_{st}(h_{dt} - h_{sd}) \tag{7}$$

$$Q_{st} = Q_{st.SHSHE} + Q_{st.PCMHE} \tag{8}$$

The sensible heat and latent heat is reused to heat the bypassed feed water and condensate water, respectively. The heat release power of the SHRHE and the PCMHE, and the total heat release power are calculated according to Eqs. (9)–(11).

$$Q_{re.SHRHE} = G_{f,by}(h_{w1} - h_{w4}) \tag{9}$$

$$Q_{re.PCMHE} = G_{c,by}(h_{w5} - h_{w8}) \tag{10}$$

$$Q_{re} = Q_{re.SHRHE} + Q_{re.PCMHE} \tag{11}$$

Due to the sensible heat storage time is equal to the latent heat storage time in TES process, so the sensible heat release time should be equal to the latent heat release time in TER process. The ratio of sensible heat release power to total heat release power should be equal to the ratio of sensible heat storage power to total heat storage power. Therefore, the sensible heat storage power factor (θ) is defined as Eq. (12).

$$\theta = \frac{Q_{st.SHSHE}}{Q_{st}} = \frac{Q_{re.SHRHE}}{Q_{re}} \tag{12}$$

The relationship between heat storage time and heat release time is calculated according to Eq. (13) and Eq. (14).

$$Q_{st.SHSHE}\tau_{st}\eta_{st}\eta_{re}\varepsilon = Q_{re.SHRHE}\tau_{re} \tag{13}$$

$$Q_{st.PCMHE}\tau_{st}\eta_{st}\eta_{re}\varepsilon = Q_{re.PCMHE}\tau_{re} \tag{14}$$

where, $\eta_{st} = 98\%$, $\eta_{re} = 98\%$, $\varepsilon = 95\%$.

3.3. Heat balance equation for operational flexibility system of CFPP

Due to the mass flow rate of each stages of the turbine changes caused by the main steam extraction, the off-design condition of the stage can be calculated by Eq. (15) [25].

$$\frac{G_{Ts,i}}{G_{Ts,0i}} = \sqrt{\frac{p_{Ts,i}^2 - p_{Ts,i+1}^2}{p_{Ts,0i}^2 - p_{Ts,0i+1}^2}}\sqrt{\frac{T_{Ts,i}}{T_{Ts,0i}}} \tag{15}$$

where, i is the number of turbine stages ($i = 1,2,\cdots,7$).

According to Eqs. (1)–(4), Eqs. (6)–(11) and Eq. (15), the heat balance equation of operational flexibility system of CFPP is obtained, as shown in Eq. (16).

$$\begin{bmatrix} q_1 & & & & & & \\ \gamma_2 & q_2 & & & & & \\ \gamma_3 & \gamma_3 & q_3 & & & & \\ \gamma_4 & \gamma_4 & \gamma_4 & q_4 & & & \\ & & & & q_5 & & \\ & & & & \gamma_6 & q_6 & \\ & & & & \gamma_7 & \gamma_7 & q_7 \end{bmatrix}\begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \\ G_5 \\ G_6 \\ G_7 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ G_{st}(h_{sd} - h_{w4}) \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} (G_f - G_{f,by})\varphi_1 \\ (G_f - G_{f,by})\varphi_2 \\ (G_f - G_{f,by})\varphi_3 \\ G_c\varphi_4 \\ (G_c - G_{c,by})\varphi_5 \\ (G_c - G_{c,by})\varphi_6 \\ (G_c - G_{c,by})\varphi_7 \end{bmatrix} \tag{16}$$

During the load reduction process, G_{st} is a variable, $G_{f,by}$ and $G_{c,by}$ are 0.

During the load raising process, G_{st} is 0, $G_{f,by}$ and $G_{c,by}$ are variables.

3.4. Heat balance equation of boiler

Due to the small fluctuation of the boiler load during the load reduction and raising process, it is necessary to recalculate the heat balance of boiler. The endothermic heat balance and combustion heat balance equations in the boiler are Eq. (17).

$$G_{coal}\psi\eta_b = G_s(h_s - h_1) + G_r(h_r - h_2) \tag{17}$$

where, $\psi = 29270$ kJ/kg, $\eta_b = 93.36\%$.

3.5. Thermal efficiency of operational flexibility system

The load reduction of CFPP provides more space for renewable energy into the grid. It is considered that the reduced power output of CFPP is equivalent to the power output of renewable energy into the grid. The equivalent power output of renewable energy into the grid can be calculated according to Eq. (18). After completing the load reduction and raising process. The total electrical power into the grid is divided into the power output of CFPP in load reduction process, the equivalent power output of renewable energy into the grid, and the power output of CFPP in load raising process. In order to evaluate the thermal efficiency of operational flexibility system, the cycle thermal efficiency and equivalent thermal efficiency are defined. The cycle thermal efficiency is the ratio of the power output in load reduction and raising process to the combustion heat of coal. The cycle thermal efficiency is calculated according to Eq. (19). The equivalent thermal efficiency is the ratio of the total electrical power into the grid to the combustion heat of coal when the CFPP completes the load reduction and raising process, which is calculated according to Eq. (20).

$$W_{st.eq} = W_{base} - W_T \tag{18}$$

$$\eta_{cycle} = \frac{W_{st.T}\tau_{st} + W_{re.T}\tau_{re} - W_{st.pu}\tau_{st} - W_{re.pu}\tau_{re}}{\psi(G_{st.coal}\tau_{st} + G_{re.coal}\tau_{re})} \times 100\% \tag{19}$$

$$\eta_{eq} = \frac{W_{st.T}\tau_{st} + W_{re.T}\tau_{re} + W_{st.eq}\tau_{st} - W_{st.pu}\tau_{st} - W_{re.pu}\tau_{re}}{\psi(G_{st.coal}\tau_{st} + G_{re.coal}\tau_{re})} \times 100\% \tag{20}$$

where, $W_{st.eq}$ is the equivalent power output of renewable energy into the grid, MW, W_{base} is the basic power output of CFPP, which is 300.03 MW in 50% THA and 360.07 MW in 60% THA, η_{cycle} is cycle thermal efficiency, %, η_{eq} is equivalent thermal efficiency, %, $G_{st.coal}$ is mass flow rate of coal in load reduction process, t/h, $G_{re.coal}$ is mass flow rate of coal in load raising process, t/h.

3.6. Calculation method for operational flexibility of CFPP

When calculating the operational flexibility of CFPP, it should be assumed that:

(1) The steam turbine and boiler operates in steady state.

(2) The steam extraction temperature of each stage remains unchanged before and after the off-design condition.

(3) The steam leakage of shaft seal and auxiliary steam consumption are ignored.

(4) The TES and TER are considered to be an ideal modal, which can meet the temperature requirements of each state point.

(5) The boiler maintains the lowest stable combustion load in load reduction process.

(6) Pressure loss is not considered during TES and TER process.

The calculation flow chart of the operational flexibility system is shown in Fig. 3. Firstly, the thermal parameters of 100% THA, 60% THA and 50% THA should be obtained according to the off-design method. Secondly, the power output of CFPP is calculated in load reduction process. Finally, the power output of CFPP with main steam parameters of under 50% and 60% THA is calculated in load raising process.

The research scheme and the parameters of extracted steam is shown in Table 2.

4. Results and discussion

4.1. Model validation

According to Fig. 3, the parameters of CFPP at 100% THA, 60% THA and 50% THA are obtained. The errors between designed value and simulation value is analyzed. As shown in Table 3. The maximum error is reheat steam pressure under 50% THA, which is 2.11%. The changes of reheat steam pressure has little effect on the enthalpy of reheat steam. It can be ignored. Other errors are less than 2%, which meets the engineering accuracy requirements. So this model [26] can be used to investigate the thermal characteristics of operational flexibility system of CFPP.

Table 2
Research scheme and parameters of extracted steam.

Research scheme	Units	I	II	III	IV
p_{st}	MPa	0.7	0.9	0.9	0.9
T_{st}	K	769.42	770.43	770.43	770.43
G_{st}	t/h	0–250	0–250	0–250	0–250
p_{dt}	MPa	0.7	0.9	0.9	0.9
T_{dt}	K	453.15	453.15	473.15	453.15
G_{dt}	t/h	0–250	0–250	0–250	0–250
p_{sd}	MPa	0.7	0.9	0.9	0.9
T_{sd}	K	433.15	433.15	433.15	428.15
G_{sd}	t/h	0–250	0–250	0–250	0–250

4.2. Effect of heat storage power, power output and equivalent power into the grid

The heat storage power, power output and equivalent power into the grid are obtained in different parameters of extracted steam, which is

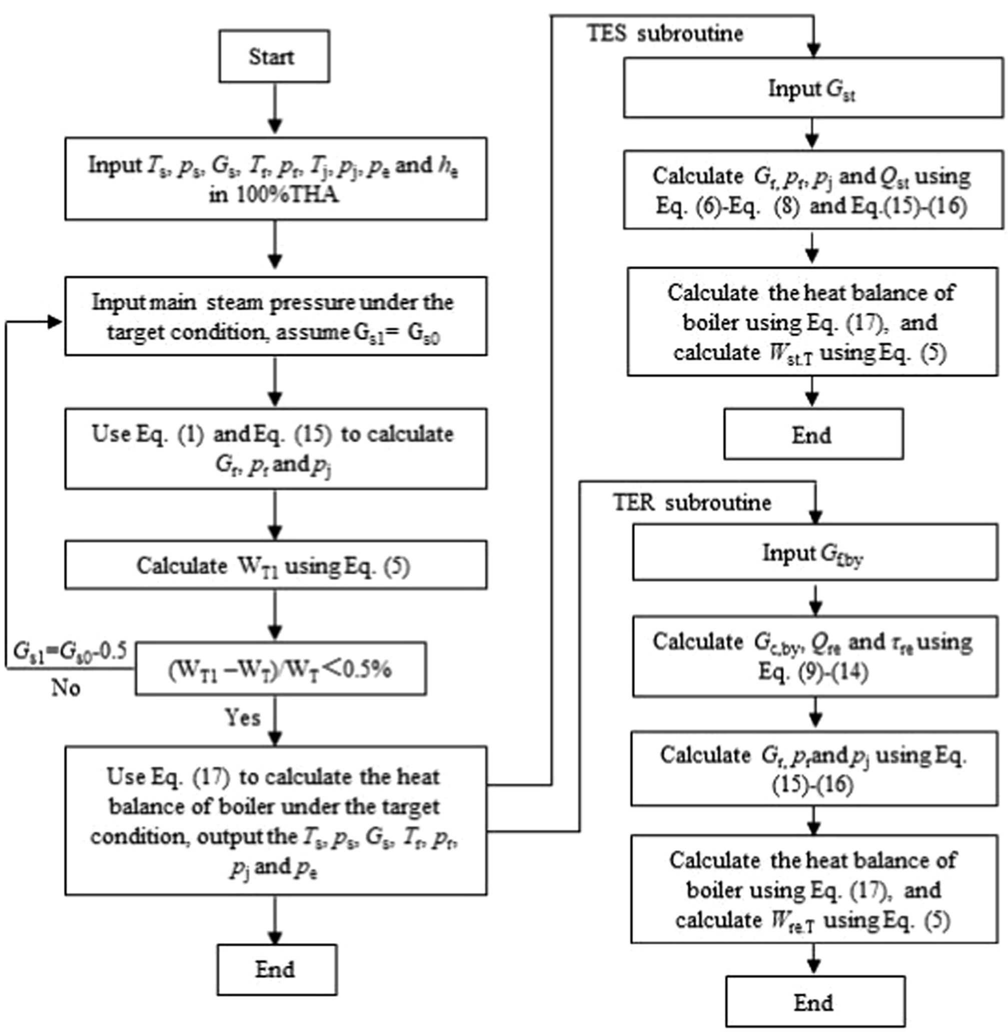

Fig. 3. Calculation flow chart for operational flexibility of CFPP.

Table 3
Error analysis.

	W_T (MW)	p_s (MPa)	G_s (t/h)	p_r (MPa)	G_r (t/h)	G_e (t/h)
100% THA						
Designed value	600.18	16.67	1848.8	3.414	1576.1	1218.3
Simulation value	600.18	16.67	1843.6	3.414	1579	1203.2
Errors (%)	0	0	0.3	0	−0.2	1.2
60% THA						
Designed value	360.13	11.61	1077.21	2.071	947.49	770.3
Simulation value	360.07	11.61	1080.8	2.062	963	768.83
Errors (%)	0	0	−0.33	0.43	−1.64	0.19
50% THA						
Designed value	300.11	9.77	900.11	1.75	798.27	659.03
Simulation value	300.03	9.77	894.3	1.713	804.23	652.57
Errors (%)	0	0	0.65	2.11	−0.75	0.98

shown in Table 4.

It can be seen from schemes (I) and (II) that the extracted steam pressure is depressurized by equal enthalpy at the TV. The pressure has little effect on the enthalpy of extracted steam condensed into pressured water in PCMHE. The mass flow rate of extracted steam of RH4 is almost unchanged after the pressured water enters into RH4. So the power output of CFPP is almost unchanged, and the heat storage power has no obvious change. From schemes (II) and (III), when the pressure of extracted steam and the temperature of drainage water are fixed, the change of dividing temperature of sensible heat and latent heat only affects the sensible heat storage power and latent heat storage power, it makes no contribution to the power output of CFPP and the heat storage power. The scheme (II) and (IV) indicates that the enthalpy of drainage water in PCMHE decreases with the decline of its outlet temperature, which leads to the raising heat storage power of extracted steam. At the same time, as the enthalpy of drainage water in PCMHE decreases, the mass flow rate of extracted steam of RH4 raises, which reduces the mass flow rate of steam into the turbine. The power output of CFPP is decreased, and the heat storage power is increased. Therefore, the lower limit of the load of CFPP is determined by the mass flow rate of extracted steam during the load reduction process. When the mass flow rate of extracted main steam of schemes (I) - (IV) is 250 t/h, the power out of CFPP is 204.5 MW, 204.51 MW, 204.51 MW and 204.21 MW respectively, and the equivalent power output of renewable energy into grid is 95.53 MW, 95.52 MW, 95.52 MW and 95.82 MW, respectively.

4.3. Effect of bypassed feed water and condensate water

The sensible heat storage power factor is determined by the pressure of extracted steam behind the TV, the dividing temperature of sensible heat and latent heat of extracted steam, and the temperature of drainage water of extracted steam. The sensible heat storage power factor affects the mass flow rate of bypassed pressured water during load raising process. From Eqs. (6)–(12), the sensible heat storage power factor of schemes (I) - (IV) are 0.2411, 0.2462, 0.2288 and 0.2443, respectively. As shown in Fig. 4.

Due to the different pressure of extracted steam behind the TV in schemes (I) and (II), the enthalpy of steam at 0.7 MPa and 453.15 K is higher than that at 0.9 MPa and 453.15 K. The sensible heat storage power of scheme (I) is less than that of scheme (II), which leads to $\theta_{II} > \theta_I$. The dividing temperature of sensible heat and latent heat is different in schemes (II) and (III). The higher dividing temperature reduces sensible heat storage power. So that $\theta_{II} > \theta_{III}$. The different parameters of scheme (II) and (IV) is the temperature of drainage water of extracted steam. The lower temperature of drainage water of extracted steam increases the total heat storage power. The proportion of sensible heat storage power in scheme (II) is less than that of in scheme (IV). So that $\theta_{II} > \theta_{IV}$.

The sensible heat storage power factor mainly affects the maximum mass flow rate of bypassed feed water in load raising process. So as to affecting the mass flow rate of condensate water. Fig. 5 shows the effect of sensible heat storage power factor on the mass flow rate of bypassed pressured water. With the increasing of mass flow rate of bypassed feed water, the mass flow rate of bypassed condensate water increases. The maximum mass flow rate of bypassed feed water differs with different the sensible heat storage power factor and the unit load. When the sensible heat storage power factors is 0.2411, 0.2462, 0.2288 and 0.2443, the corresponding maximum mass flow rate of bypassed feed water is 243.5 t/h, 251 t/h, 225.85 t/h and 248.25 t/h, respectively, under the main steam parameters of 50% THA during the load raising

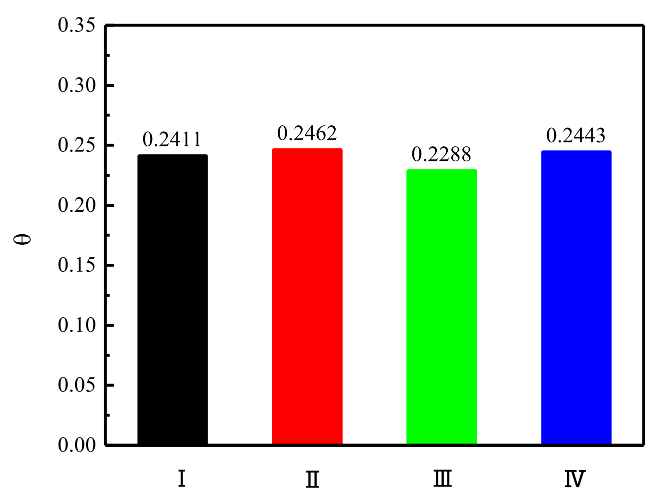

Fig. 4. Sensible heat storage power factor of schemes (I) - (IV).

Table 4
Q_{st}, $W_{st,T}$ and $W_{st,eq}$ in different parameters of extracted steam.

Research scheme	G_{st} (t/h)	0	50	100	150	200	250
I	Q_{st} (MW)	0	38.87	77.74	116.61	155.48	194.36
	$W_{st,T}$ (MW)	300.03	280.61	261.36	242.31	223.15	204.5
	$W_{st,eq}$ (MW)	0	19.42	38.67	57.72	76.88	95.53
II	Q_{st} (MW)	0	38.87	77.74	116.61	155.48	194.34
	$W_{st,T}$ (MW)	300.03	280.57	261.36	242.31	223.15	204.51
	$W_{st,eq}$ (MW)	0	19.46	38.67	57.72	76.88	95.52
III	Q_{st} (MW)	0	38.87	77.74	116.61	155.48	194.35
	$W_{st,T}$ (MW)	300.03	280.57	261.36	242.31	223.15	204.51
	$W_{st,eq}$ (MW)	0	19.46	38.67	57.72	76.88	95.52
IV	Q_{st} (MW)	0	39.17	78.34	117.51	156.68	195.85
	$W_{st,T}$ (MW)	300.03	280.54	261.22	241.82	222.92	204.21
	$W_{st,eq}$ (MW)	0	19.49	38.81	58.21	77.11	95.82

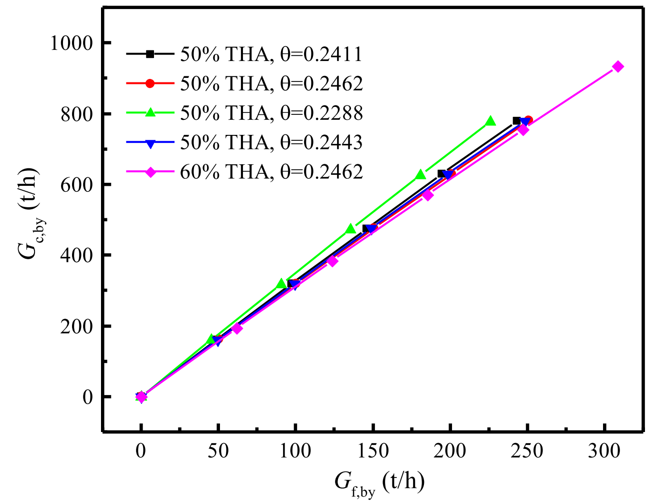

Fig. 5. Effect on bypassed feed water and condensate water.

process. While the condensate water flows into RH7 is 0. When the unit is running at the condition with the main steam parameters of 60% THA, and the sensible heat storage power factor is 0.2463, the maximum mass flow rate of bypassed feed water is 308.65 t/h.

4.4. Effect of power output of CFPP during load raising process

During the load raising process, the increment of power output of CFPP is defined ($\triangle W_{re.T}$), according to Eq. (5).

$$\Delta W_{re.T} = W_{re.T} - W_{base} \tag{21}$$

The effect of the sensible heat storage power factor on the increment of power output during the load raising process is obtained, as shown in Fig. 6. The increasing mass flow rate of bypassed feed water results in the raising of increment of power output of CFPP. The reason is that the increasing mass flow rate of bypassed feed water leads to the decrease of the mass flow rate of pressured water into RH1 - RH3 and RH5 - RH7. The mass flow rate of extracted steam of RH1 - RH3 and RH5 - RH7 is reduced. So the mass flow rate of steam into the turbine increases. Then the increment of power output of CFPP raises. The larger sensible heat storage power factor leads to the bigger maximum mass flow rate of the bypassed feed water and condensate water, the higher maximum increment of power output of CFPP can be achieved. When the sensible heat storage power factors is 0.2411, 0.2462, 0.2288 and 0.2443, the maximum increment of power output of CFPP is 25.21 MW, 25.44 MW, 24.29 MW and 25.4 MW, respectively, under the main steam parameters of 50% THA during the load raising process. When the unit is running at the condition with the main steam parameters of 60% THA, and the sensible heat storage power factor is 0.2463, the maximum increment of power output of CFPP is 34.13 MW. Which is greater than that under 50% THA.

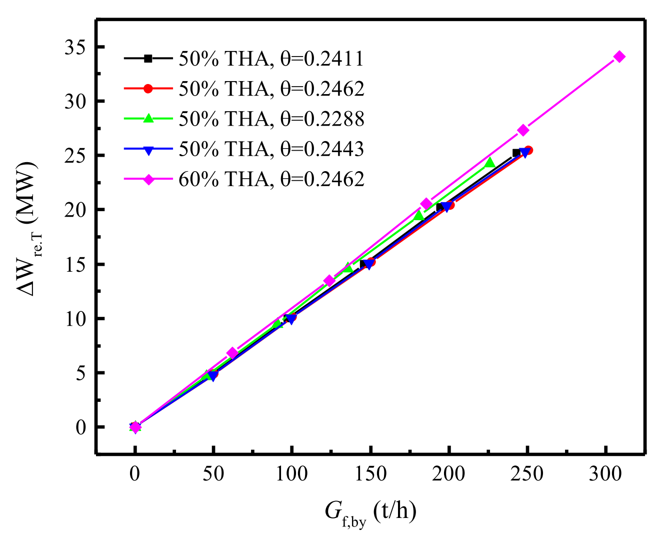

Fig. 6. Effect of sensible heat storage power factor on increment of power output of CFPP.

According to Table 4 and Fig. 6, the load range of CFPP is obtained during load reduction and raising process. As shown in Table 5. The minimum power output of schemes (I) - (Ⅳ) is 204.5 MW, 204.51 MW, 204.51 MW and 204.21 MW under the main steam parameters of 50% THA during the load reduction process, respectively. The maximum power output of CFPP is 325.24 MW, 325.47 MW, 324.32 MW and 325.43 MW under the main steam parameters of 50% THA during the load raising process, respectively. And when the sensible heat storage power factor is 0.2462, the maximum power output is 394.2 MW under the condition with the main steam parameters of 60% THA during the load raising process.

It can be seen from Table 5 and Fig. 6, the maximum increment of power output can be achieved at the maximum of the sensible heat storage power factor. Under the condition with the maximal sensible heat storage power factor of 0.2462 and the main steam parameters of 50% THA during the load reduction and raising process, the unit load can be widen from 34.07 – 54.23% THA. With the same sensible heat storage power factor and the load reduction process, the unit load can be widen from 34.07 – 65.68% THA under the condition with the main steam parameters of 60% THA during the load raising process.

4.5. Effect of heat release time

The heat release time determines the deep peak shaving time of CFPP. When the heat release time is calculated, 194.34 MWh is selected as the heat storage capacity according to the scheme (II) with the mass flow rate of extracted steam of 250 t/h and the heat storage time of 1 h. Fig. 7 shows the heat release time. Corresponding to the sensible heat storage power factor of scheme (I) - (Ⅳ) , the shortest heat release time is 1.87 h, 1.86 h, 1.91 h and 1.86 h under the main steam parameters of 50% THA during load raising process, respectively. It is found that when the maximum sensible heat storage power factor is 0.2462, the peak shaving capacity of CFPP is the largest. The operational flexibility of CFPP can meet the requirement of deep peak shaving of 8.4 h/day. When the sensible heat storage power factor is 0.2462 under the condition with the main steam parameters of 60% THA during the load raising process, the heat release time is 1.43 h. The unit can meet the requirement of deep peak shaving of 9.86 h/day.

Table 5
Load range of CFPP during load reduction and raising process (load reduction process is at the main steam parameters of 50% THA).

	Load raising process at main steam parameters of 50% THA				Load raising process at main steam parameters of 60% THA
θ	0.2411	0.2462	0.2288	0.2443	0.2462
$W_{st,T}$ (MW)	204.5	204.51	204.51	204.21	204.51
Minimum load (% THA)	34.07	34.07	34.07	34.02	34.07
$W_{re,T}$ (MW)	325.24	325.47	324.32	325.43	394.2
Maximum load (% THA)	54.19	54.23	54.04	54.22	65.68

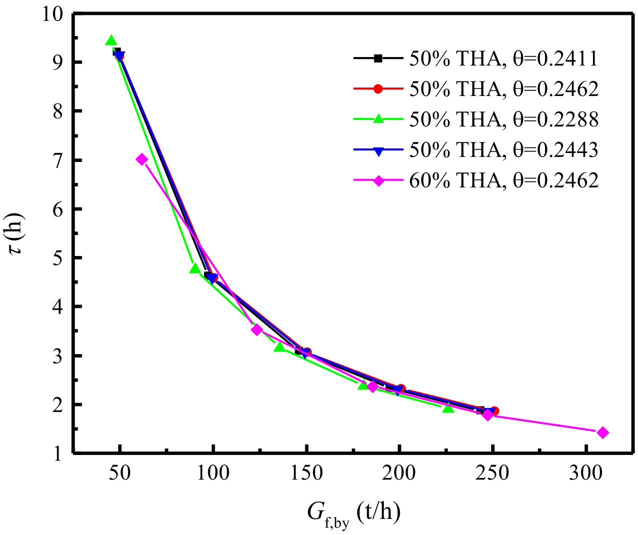

Fig. 7. Heat release time.

4.6. Effect of thermal efficiency

The heat storage capacity and work condition described in 4.5 shall be selected. Fig. 8 shows the relationship between the cycle thermal efficiency and operating conditions, where is no renewable energy entered into the grid. The additional heat loss and increasing heat rate of the unit caused by bypassed feed water and condensate water leads to the decline of cycle thermal efficiency. So, with the increasing mass flow rate of bypassed feed water, the cycle thermal efficiency of CFPP decreases. When the load is running at the condition with main steam parameters of 50% THA during load raising process, and the sensible heat storage power factors is 0.2411, 0.2462, 0.2288 and 0.2443 respectively, the cycle thermal efficiency at the maximum mass flow rate of bypassed feed water is 36.12%, 36.11%, 36.12% and 36.13%. Compared with 50% THA of 38.18%, the cycle thermal efficiency is decreased by 2.06%, 2.07%, 2.06% and 2.05% respectively. When the sensible heat storage power factor is 0.2462, and the unit is running at the condition with main steam parameters of 60% THA during load raising process, the cycle thermal efficiency is 36.44% at the maximum mass flow rate of bypassed feed water. Compared with the 60% THA of 38.51%, it is reduced by 2.07%. Due to the heat rate of CFPP in 60% THA is higher than that in 50% THA, the cycle thermal efficiency at the condition with main steam parameters of 60% THA during load raising process is bigger than that in 50% THA.

Although the heat loss and the increasing heat rate of CFPP results in the decline of cycle thermal efficiency, the increasing renewable energy into the grid in load reduction process makes the total electrical power into the grid rise. The equivalent thermal efficiency of the energy system is improved. Fig. 9 shows the relationship between the equivalent thermal efficiency of CFPP and the operating conditions, where has renewable energy entered into the grid. It can be seen that the equivalent thermal efficiency of CFPP increases with the raising of mass flow rate of bypassed feed water. The reason is that the changed power output of CFPP caused by the raising of mass flow rate of bypassed feed water is greater than coal consumption caused by the changes of feed water and reheated steam. The larger mass flow rate of bypassed feed water results in the higher the ratio of increment of power output to coal consumption. So the equivalent thermal efficiency increases. When the load is running at the condition with main steam parameters of 50% THA during load raising process, and the sensible heat storage power factors is 0.2411, 0.2462, 0.2288 and 0.2443 respectively, the equivalent thermal efficiency at the maximum mass flow rate of bypassed feed water is 40.45%, 40.46%, 40.39% and 40.46%. Compared with the reference condition of 38.18%, the equivalent thermal efficiency is increased by 2.27%, 2.28%, 2.21% and 2.28%, respectively. When the sensible heat storage power factor is 0.2462, and the unit is running at the condition with main steam parameters of 60% THA during load raising process, the equivalent thermal efficiency is 41.06% at the maximum mass flow rate of bypassed feed water. Compared with the reference condition of 38.51%, it is increased by 2.55%.

5. Conclusion

The 600 MW subcritical coal-fired power plant with 50% rated load (300.03 MW) is selected as the research model to adjust the unit load. The integrating of the main steam extraction system with coal-fired power plant to realize the operational flexibility is studied. The impacts of the extracted steam parameters and the operation load on the coal-fired power plant are analyzed during the load reduction and raising process. The main conclusions are listed as follows:

(1) The novel main steam extraction system integrated with coal-fired power plant can achieve the operational flexibility when the unit operates in the lowest stable load. The power output of coal-fired power plant can be reduced to 204.51 MW from 300.03 MW when the main steam extraction mass flow rate is 250 t/h during the load reduction process. When the maximal sensible heat storage power factor is 0.2462

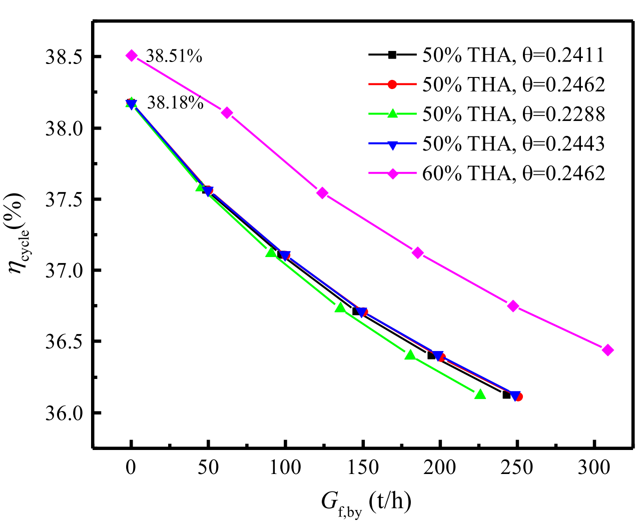

Fig. 8. Relationship between cycle thermal efficiency of CFPP and operating conditions.

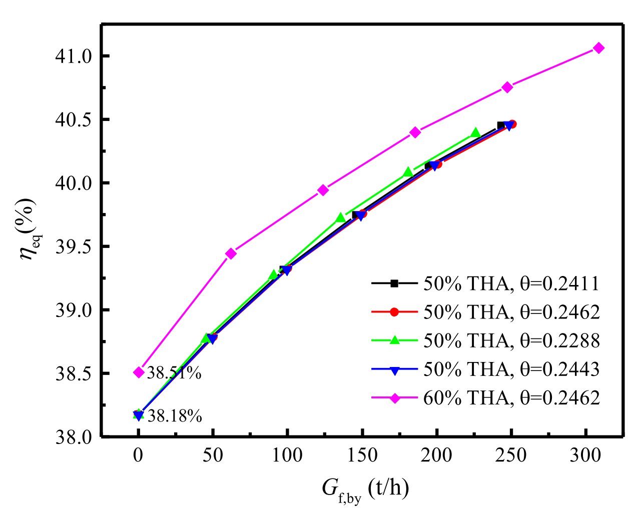

Fig. 9. Relationship between equivalent thermal efficiency of CFPP and operating conditions.

and the mass flow rate of the bypassed feed water is 251 t/h, the power output of coal-fired power plant can be increased to 325.47 MW from 300.03 MW during the load raising process. The unit load range can be widen from 34.07 to 54.23%. When the maximal sensible heat storage power factor is 0.2462 and the mass flow rate of the bypassed feed water is 308.65 t/h, the power output of coal-fired power plant can be increased to 394.2 MW from 360.07 MW during the load raising process. The unit load range can be widen from 34.07 to 65.68%.

(2) When the operational flexibility of the coal-fired power plant is completed with the condition that load reduction by 204.51 MW and load raising by 325.47 MW, the coal-fired power plant achieves the maximum deep peak shaving time of 8.4 h/d and increases the equivalent thermal efficiency by 2.28%.

(3) When the operational flexibility of the coal-fired power plant is completed with the condition that load reduction by 204.51 MW and load raising by 394.2 MW, the coal-fired power plant achieves the maximum deep peak shaving time of 9.86 h/d and increases the equivalent thermal efficiency by 2.55%.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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